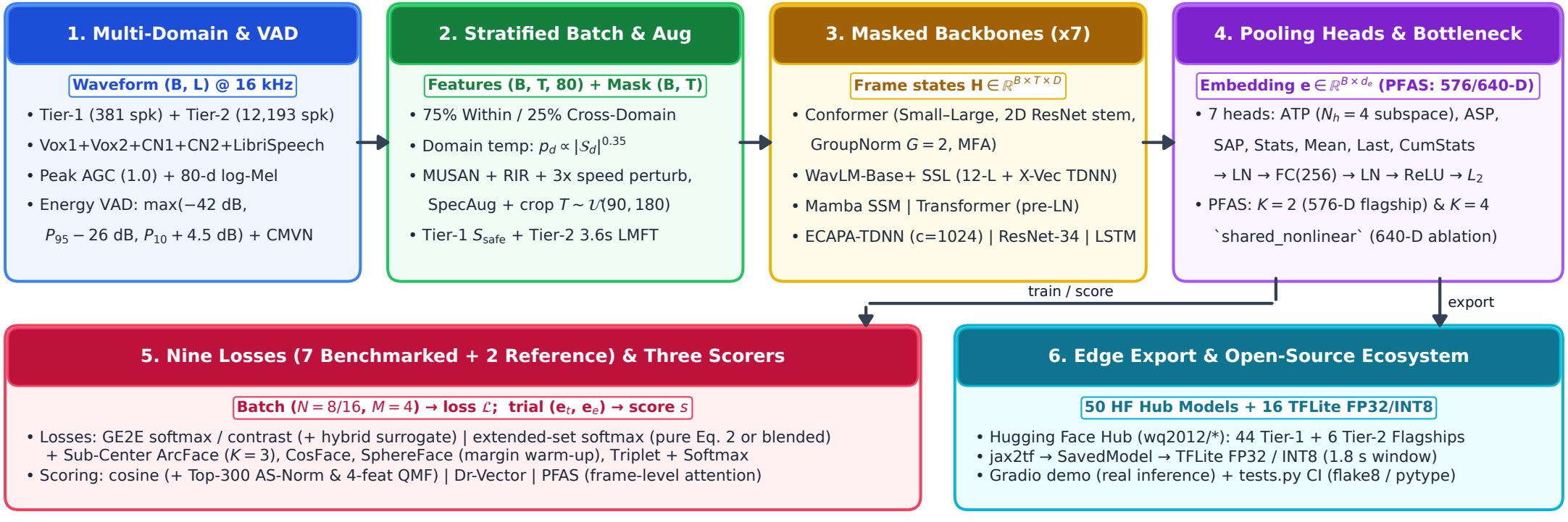

FlaxSpeaker: Multi-Domain Speaker Verification Pipeline (JAX / Flax)
1. Multi-Domain & VAD
Waveform (B, L) @ 16 kHz
• Tier-1 (381 spk) + Tier-2 (12,193 spk)
• Vox1+Vox2+CN1+CN2+LibriSpeech
• Peak AGC (1.0) + 80-d log-Mel
• Energy VAD: max(−42 dB, $P_{95} - 26$ dB, $P_{10} + 4.5$ dB) + CMVN
2. Stratified Batch & Aug
Features (B, T, 80) + Mask (B, T)
• 75% Within / 25% Cross-Domain
• Domain temp: $p_d \propto |S_d|^{0.35}$
• MUSAN + RIR + 3x speed perturb, SpecAug + crop $T \sim \mathcal{U}(90, 180)$
• Tier-1 S_{safe} + Tier-2 3.6s LMFT
3. Masked Backbones (x7)
Frame states H $\in \mathbb{R}^{B \times T \times D}$
• Conformer (Small–Large, 2D ResNet stem, GroupNorm $G = 2$, MFA)
• WavLM-Base+ SSL (12-L + X-Vec TDNN)
• Mamba SSM | Transformer (pre-LN)
• ECAPA-TDNN (c=1024) | ResNet-34 | LSTM
4. Pooling Heads & Bottleneck
Embedding e $\in \mathbb{R}^{B \times d_e}$ (PFAS: 576/640-D)
• 7 heads: ATP ($N_h = 4$ subspace), ASP, SAP, Stats, Mean, Last, CumStats → LN → FC(256) → LN → ReLU → L_2
• PFAS: $K = 2$ (576-D flagship) & $K = 4$ `shared_nonlinear` (640-D ablation)
train / score
export
5. Nine Losses (7 Benchmarked + 2 Reference) & Three Scorers
Batch ($N = 8/16$, $M = 4$) → loss $\mathcal{L}$; trial (e_t, e_e) → score s
• Losses: GE2E softmax / contrast (+ hybrid surrogate) | extended-set softmax (pure Eq. 2 or blended) + Sub-Center ArcFace ($K = 3$), CosFace, SphereFace (margin warm-up), Triplet + Softmax
• Scoring: cosine (+ Top-300 AS-Norm & 4-feat QMF) | Dr-Vector | PFAS (frame-level attention)
6. Edge Export & Open-Source Ecosystem
50 HF Hub Models + 16 TFLite FP32/INT8
• Hugging Face Hub (wq2012/*): 44 Tier-1 + 6 Tier-2 Flagships
• jax2tf → SavedModel → TFLite FP32 / INT8 (1.8 s window)
• Gradio demo (real inference) + tests.py CI (flake8 / pytype)